# *AI-Related Mergers and Labor Markets: Introducing BET to Antitrust Review*

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**Abstract**

AI-related mergers can affect both product-market competition and employer-side competition for labor. Because AI systems can automate, augment, or alter human tasks in some settings, antitrust analysis of such transactions should not be limited to relevant product markets; it should also consider relevant buyer-side markets, including labor markets in which employers purchase labor. The practical antitrust question is not whether AI generally helps or harms workers. Rather, it is whether a particular transaction is likely to alter competition in a relevant labor market relative to the likely but-for world.

This article introduces Binary Expansion Testing ("BET") as a useful preliminary tool for antitrust lawyers analyzing AI-related merger matters. BET tests whether a measurable indicator of AI adoption or use and a labor-market outcome behave independently or instead show a recurring, non-random pattern that warrants further investigation. The article explains BET in plain antitrust language and uses a Washington-Pennsylvania unemployment comparison as an illustration. Public state-level data can help show why common shocks, later divergence, and aggregation problems matter. In actual merger review, however, any BET-based inquiry would need to be tied to granular evidence, such as firm-level, platform-level, occupation-level, or worksite-level data connected to the asserted theory of harm. The article offers a conceptual, forward-looking framework rather than a description of current merger-review practice.

## Introduction

AI is no longer only a product-market issue. Firms use AI systems to screen resumes, rank applicants, forecast demand, write code, draft content, manage logistics, allocate tasks, monitor workers, schedule shifts, price products, automate support, and assist creative or analytical work. A merger involving AI systems, AI talent, cloud infrastructure, data assets, or AI-enabled workflow tools may therefore affect both downstream product markets and upstream labor markets.

That possibility alone does not establish a Section 7 concern. Section 7 of the Clayton Act asks whether the effect of an acquisition may be substantially to lessen competition or to tend to create a monopoly. The 2023 Merger Guidelines frame merger review as a forward-looking inquiry into whether a transaction may lessen competition, including competition among employers as buyers of labor. The inquiry is transaction-specific: a merger should not be challenged because AI is controversial, and it should not be allowed merely because AI may produce efficiencies.[1]

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[1] Clayton Act Section 7, 15 U.S.C. § 18; U.S. Department Of Justice ("DoJ") & Federal Trade Commission ("FTC"), Merger Guidelines (*hereafter* "Merger Guidelines"), Sections 1, 2.10, 4.3.D.8 (2023), https://www.justice.gov/atr/2023-merger-guidelines.

AI-related labor effects can be indirect, nonlinear, and uneven. They are indirect when the effect appears through task redesign, fewer vacancies, algorithmic scheduling, worker monitoring, or control over data that changes bargaining conditions. They are nonlinear when a small amount of AI adoption has little visible effect until a threshold is crossed. They are uneven when the same tool affects junior engineers, call-center employees, warehouse workers, artists, analysts, or retail workers differently.

The legal problem is causation. Courts, agencies, and counsel must ask whether the transaction, rather than AI in the abstract, is likely to reduce competition in a relevant market. Statistical tools alone cannot answer that legal question. BET is a test of statistical independence; here, it is used to ask a narrower preliminary question: do the tested variables behave as if they are unrelated, or do they show a recurring, non-random relationship that may justify further factual and economic inquiry?[2]

The article proceeds in four parts. Part I defines the scope of AI-related merger review and explains why worker substitution is context-specific. Part II explains why labor-market harm must be merger-specific. Part III introduces BET in plain antitrust language and explains how it differs from correlation, regression, and causal proof. Part IV uses the Washington-Pennsylvania unemployment comparison as an illustration.

## I. Scope and Definitions

### A. What Counts as AI and an AI-related Merger

For purposes of this article, AI means a machine-based system that uses data, rules, statistical models, or learned parameters to infer outputs such as predictions, rankings, classifications, recommendations, content, or decisions that can influence physical or virtual environments. This formulation tracks the OECD definition of an AI system, which emphasizes machine-based inference, outputs, autonomy, and possible adaptiveness after deployment.[3]

The definition includes large language models, generative AI systems, recommender systems, predictive analytics, computer-vision tools, AI-enabled workflow systems, and algorithmic-management tools when those systems are relevant to how the transaction may affect competition, labor-market outcomes, innovation, or market structure. Traditional machine-learning techniques can fall within the definition when they use data and inference to generate predictions or decisions. A purely deterministic software tool that follows fixed rules without inference, prediction, ranking, or adaptation is not the focus of this article, although it may matter in merger review for other reasons.

The definition also reflects a practical labor-market point. AI often affects tasks rather than entire jobs. A job may combine writing, coding, analysis, communication, scheduling, judgment, and physical work. AI may automate or augment some tasks while leaving others unchanged. For

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[2] Kai Zhang, *BET on Independence*, 114 JOURNAL OF THE AMERICAN STATISTICAL ASSOCIATION 1620 (2019), *available at* https://doi.org/10.1080/01621459.2018.1537921.

[3] OECD, Explanatory Memorandum on the Updated OECD Definition of an AI System, OECD Artificial Intelligence Papers No. 8 (Mar. 2024), *available at* https://doi.org/10.1787/623da898-en; OECD Recommendation of the Council on Artificial Intelligence, OECD/LEGAL/0449.

that reason, a labor-market effect may appear as slower hiring, changed skill requirements, altered promotion paths, increased monitoring, or changed job quality rather than immediate layoffs.[4]

An AI-related merger is a transaction in which the target, acquirer, or assets being combined materially involve AI technologies, AI inputs, AI deployment, or AI-enabled labor-management practices. The category is broader than acquisitions of foundation-model companies. It may include transactions involving specialized chips, cloud infrastructure, proprietary data, AI applications, enterprise workflow tools, algorithmic-management tools, or teams whose specialized AI talent is itself a competitive asset.

### B. Labor Markets Are Dynamic

Workers can learn new skills, relocate, switch employers, and enter or exit occupations. That makes labor markets dynamic. But dynamism does not make all workers substitutes for all other workers. A grocery cashier, a machine-learning researcher, a data-center electrician, and a storyboard artist may all be affected by AI, but they do not usually compete for the same jobs or constrain the same employers' hiring decisions. The antitrust question is whether employers compete for the same workers, or for workers who can realistically switch into the relevant jobs within a practical time frame.

### C. Levels of Evidence

State-level evidence occupies a limited place in this framework. State unemployment, industry composition, vacancies, wage growth, job postings, and occupational shifts rarely prove merger-specific harm. Too many factors affect state labor markets, including macroeconomic shocks, demographics, state law, demand conditions, industry cycles, migration, interest rates, weather, and public policy.

Public data can help generate testable hypotheses at an early stage. If two states experience a common shock, such as a public-health emergency, but later diverge in a way that aligns with differences in technology exposure, industry composition, or adoption timing, the pattern may warrant further inquiry. In a merger case, any such hypothesis must be narrowed to the relevant labor market and tested with granular evidence. Public data should not be treated as proof that the transaction caused observed labor harm; instead, they can help antitrust enforcement agencies ask whether state-specific factors or the transaction better explain the pattern.

## II. Labor-Market Harm Must Be Merger-Specific

### A. Labor-Market Competition Is an Antitrust Concern

Labor-market competition is a recognized antitrust concern. The 2023 Merger Guidelines define labor markets as markets "where employers are the buyers of labor and workers are the sellers" and explain that labor markets are important buyer markets.[5] When a merger combines competing buyers, including employers, the agencies examine whether the transaction may substantially lessen competition for workers, creators, suppliers, or other providers.[6]

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[4] Tyna Eloundou, Sam Manning, Pamela Mishkin & Daniel Rock, *GPTs Are GPTs: Labor Market Impact Potential of LLMs*, 384 SCIENCE 1306 (2024).

[5] DoJ & FTC, Merger Guidelines Section 2.10, at 26–27 (Dec. 18, 2023).

[6] MERGER GUIDELINES Section 2.10 and Guideline 10 (2023); DOJ & FTC, ANTITRUST GUIDELINES FOR BUSINESS ACTIVITIES AFFECTING WORKERS 1–3 (Jan. 16, 2025), *available at* https://www.ftc.gov/system/files/ftc_gov/pdf/p251201antitrustguidelinesbusinessactivitiesaffectingworkers2025.pdf.

That point is consistent with established antitrust case law. The Supreme Court has long recognized that antitrust law protects sellers injured by buyer-side restraints, and workers are sellers of labor.[7] Modern merger matters likewise recognize that transactions can affect competition among employers, including competition for authors, retail workers, healthcare workers, engineers, and other labor inputs.[8]

This does not convert every labor issue into an antitrust issue. A bad workplace, a controversial automation plan, or a general decline in hiring is not an antitrust issue. The competition question is whether the merger changes the choices available to workers or changes the ability of employers to compete for those workers.

### B. The Merger-Specific Question

Merger review examines the marginal effect of the transaction. A proposed merger may be lawful even if AI is causing broad labor-market disruption. Conversely, a merger may raise labor-market concerns even if AI generally improves productivity. The question is whether this transaction changes competition among employers, platforms, or input owners in a way that may make workers worse off compared with the likely but-for world.

The practical legal question is simple: what can the combined firm do after closing that either party alone could not do, or would be less likely to do, absent the transaction? In labor markets, possible answers include reducing bidding for the same workers, centralizing wage or scheduling decisions, acquiring a tool that reduces demand for a category of workers, controlling an input used by rival employers, acquiring workforce data that changes bargaining conditions, or integrating a platform rule that makes it harder for workers to switch among, or work simultaneously through, competing employers or platforms.

A general claim that AI may reduce employment in some occupations is therefore insufficient. Such a claim requires evidence that the parties compete for the same workers, that the transaction changes workers' outside options, that the acquired AI asset gives the merged firm new power over hiring or wage-setting, or that the transaction changes access to an AI input in a way that affects labor demand in a relevant market.

### C. Industry Context Matters

Labor-market theories of harm differ across industries. In grocery, healthcare, entertainment, transportation, and local services, concerns may be direct because employers compete for overlapping workers in identifiable geographies. In software, cloud computing, semiconductors, and transactions involving foundation models or large language models, labor effects may be most salient for specialized engineers, researchers, data-center workers, or adjacent technical occupations rather than for the general workforce. In enterprise software or algorithmic-management transactions, the theory of harm may concern tools that shape scheduling, monitoring, hiring, compensation, or job quality.

This industry-specific framing disciplines the data request. If the theory of harm concerns a loss of competition for local pharmacists, the relevant evidence will differ from the evidence needed

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[7] *Mandeville Island Farms, Inc. v. American Crystal Sugar Co.*, 334 U.S. 219, 235–36 (1948); *Weyerhaeuser Co. v. Ross-Simmons Hardwood Lumber Co.*, 549 U.S. 312, 320–26 (2007); *National Collegiate Athletic Ass'n v. Alston*, 594 U.S. 69 (2021).

[8] *United States v. Bertelsmann SE & Co. KGaA*, 646 F. Supp. 3d 1 (D.D.C. 2022); *In re* Kroger Co. & Albertsons Cos., FTC Dkt. No. 9428 (Feb. 26, 2024), Complaint, Paragraphs 74–80.

to evaluate a theory involving machine-learning researchers or warehouse workers subject to algorithmic scheduling. Such specificity can make labor-market theories of harm more demanding to frame, particularly early in an investigation, when transaction-specific data may be incomplete or unavailable. Quantitative tools should therefore be understood not as substitutes for evidence gathering but as supplementary methods that can become more useful as data become available, helping investigators test proposed mechanisms of harm and detect patterns that may otherwise be missed.

### D. Causation as a But-For Inquiry

Causation in merger review is forward-looking and counterfactual. The question is not merely whether a labor outcome changed. It is whether the transaction is likely to cause a substantial lessening of competition relative to the world that would likely exist without the merger. In labor markets, the evidence may include wages, benefits, vacancies, hiring rates, separations, worker mobility, job quality, schedules, and the availability of alternative employers.

BET can help at the beginning of that inquiry by identifying whether a measurable indicator of AI adoption and a labor outcome show a recurring, non-random pattern of dependence rather than statistical independence. That finding does not establish legal causation. But it can help determine whether the data reveal a pattern worth testing with stronger evidence, including internal documents, market-definition evidence, ordinary-course business data, worker testimony, and causal econometric analysis tied to the transaction.

## III. Introducing BET to the Antitrust Quantitative Toolbox

### A. BET in Plain Antitrust Language

BET is a statistical method for testing independence. In ordinary language, independence means that knowing one thing should not help predict the other. If AI deployment and hiring outcomes are independent, then months, teams, or worksites with higher AI use should not systematically line up with lower hiring, lower pay, greater schedule volatility, more layoffs, or other involuntary job exits. If those patterns do line up in a recurring, non-random way, BET can flag the relationship for further investigation.[9]

One can think of BET as asking repeated yes-or-no questions about the data at different levels of detail. Is AI use high or low? Is hiring up or down? Is the relevant period before or after deployment? Is a worker group above or below a threshold of exposure? If two variables are unrelated, the yes-or-no patterns in one should not systematically line up with the yes-or-no patterns in the other. BET checks for those patterns without requiring the analyst to draw one straight line through all the data.

This feature makes BET particularly useful for preliminary issue-spotting in AI-related questions. AI effects may involve thresholds, regime changes, local effects, or occupation-specific differences that a simple average correlation can flatten. BET's model-light design can detect such patterns without requiring a linear relationship, a particular data distribution, or a predetermined story about how AI must affect workers.

### B. Why Ordinary Correlation Is Not Enough

Ordinary correlation can be useful, but it is most comfortable with simple relationships. It can miss thresholds, regime changes, and mixed patterns. In the AI context, that limitation matters. A

[9] Zhang, *supra* note 2.

relationship may be positive during a common shock, weak during a recovery, and negative during a later technology-adoption period. A simple average can obscure the feature that matters for legal analysis.

Regression analysis, event studies, synthetic controls, and difference-in-differences designs can be more powerful when the facts support them. In a developed merger investigation, those tools may be necessary. BET plays a complementary role. It is best understood as a model-light dependence-testing framework that helps identify whether a recurring, non-random pattern exists in the data. If such a pattern appears, lawyers and economists can then ask whether it has a plausible transaction-specific explanation and whether stronger evidence is available to test causation.

### C. Where BET Fits Conceptually

BET fits best as an introductory issue-spotting tool. Its role is to help lawyers and economists think about whether two variables appear to behave independently or whether their relationship deserves closer attention. In an AI-related merger matter, those variables might be an AI-deployment measure and a labor-market outcome, such as hiring, vacancies, wages, hours, separations, or schedule volatility.

That focused role is important. A BET result does not define the relevant labor market, identify the but-for world, measure market power, or prove that a transaction caused harm. It also does not determine whether a merger may substantially lessen competition. Instead, BET can help organize the first empirical question: is there a recurring, non-random relationship in the data that should be investigated further using merger-specific evidence?

For that reason, the most appropriate way to understand BET in this article is as a bridge between descriptive evidence and more developed causal inquiry. It can help generate questions for documents, interviews, business testimony, and economic analysis. But it should not be treated as a substitute for those forms of evidence.

### D. A Prior Platform-level Application

A prior Antitrust Report article on TikTok/Douyin helps answer the practical question of whether BET can be applied outside a purely abstract setting. That article was not a merger case and did not study labor markets. Its relevance is methodological. The authors used BET to study platform-level Douyin download data on iOS and Android devices in the context of a platform business event. The lesson for this article is not the substantive result of that study; it is that BET can be run on firm- or platform-level data tied to a concrete business event rather than only on macroeconomic public data.[10]

The same logic should apply in merger review. If the theory concerns an AI scheduling system, the relevant data may be worksite schedules, hours, quit rates, and deployment logs. If the theory concerns an AI coding assistant, the relevant data may be team-level usage, junior-engineer hiring, job postings, starting pay, and internal headcount plans. The statistical pattern is persuasive only when it is paired with a concrete mechanism.

### E. BET's Role and Limits

BET is not a legal doctrine, and it has not been adopted by the agencies as a standard merger screening tool. This article does not argue that BET should replace ordinary merger analysis or that

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[10] Y. Yang, Z. Li, Z. Zhao, S. (Zi Tong) Peng, L. Dou, C. Li & Y. Wang, *How TikTok 'Locks In' Its Users with Reference to What Happened in the Homeland Version Douyin Application*, 3 ANTITRUST REPORT, 2-23 (Nov. 2023).

a BET result should be treated as evidence of legal causation. BET does not define the relevant labor market. It does not prove that a merger caused a labor effect. It does not establish market power, competitive effects, or the likely but-for world. It does not replace internal documents, business testimony, worker evidence, concentration analysis, or causal econometrics.

BET also does not create a safe harbor when no dependence is found. A failure to detect dependence may reflect true independence, but it may also reflect small samples, poor measurement, delayed effects, noisy public data, or the wrong level of aggregation. Even with those limits, BET gives antitrust lawyers another useful way to think about recurring, non-random relationships in data, especially when AI-related labor effects may be nonlinear, uneven, or difficult to summarize with ordinary correlation.

## IV. Washington-Pennsylvania Illustration

### A. Why this Example Is Included

The Washington-Pennsylvania unemployment comparison provides a concrete teaching illustration of BET's issue-spotting role. It is not proposed as an enforcement screen, a merger screen, or evidence that AI caused any labor-market effect. The example uses public state unemployment data because those data are accessible and familiar.[11] Firm-level AI deployment logs, payroll data, worksite schedules, and occupation-specific hiring data would be more relevant in an actual merger investigation, but they are usually not publicly available. The use of publicly available state-level unemployment data in the Washington-Pennsylvania example is intended only to illustrate how a dependence-testing method can help lawyers think about common shocks, later divergence, and the limits of aggregate data. Because these data are statewide unemployment measures, they do not identify employer-level conduct, occupation-specific labor markets, worker mobility patterns, or transaction-specific causal mechanisms. A state-level result may therefore identify a question for further investigation; it cannot define a relevant labor market, prove competitive effects, or answer the transaction-specific causation question.

With that said, publicly available state-level data can be useful at an early stage because they help guide the direction of further evidence-gathering, including the collection of firm-level data.

### B. How BET Is Used in the Illustration

The inputs are simple: one column of monthly Washington unemployment rates and one column of monthly Pennsylvania unemployment rates. The independence question is whether the two columns behave as if they are unrelated. The illustration considers the pattern by period, not as a single legal conclusion.

The useful feature is the period-by-period pattern. During the COVID shock and recovery, both states experienced high unemployment and moved together because of a common shock. Later, Pennsylvania returned closer to pre-COVID levels while Washington remained more elevated and then rose further. A panel regression with time controls and a pre/post-COVID indicator could also be informative. BET serves a complementary role by flagging changing or nonlinear dependence without requiring the analyst to select a particular functional form or breakpoint in advance.

That is the investigative lesson of the illustration. The public data do not show AI causation. They show the kind of pattern that should make a lawyer ask better questions: Which occupations

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[11] *See* U.S. Bureau of Labor Statistics, Unemployment Rate in Washington [WAUR] and Unemployment Rate in Pennsylvania [PAUR], FRED, Federal Reserve Bank of St. Louis, monthly seasonally adjusted series, updated May 7, 2026; *see also* U.S. Bureau of Labor Statistics, State Employment and Unemployment.

diverged? Which firms changed hiring? Which worksites adopted new tools? Did job postings change before or after AI deployment? Do documents describe plans to substitute software, alter schedules, or reduce hiring? Those questions, not the state-level data alone, would matter in an actual merger investigation.

### C. A Plausible Mechanism to Investigate

Industry composition supplies a plausible reason to investigate Washington more closely in some AI-related theories of harm. Washington economic-development materials describe the state as a major information and communication technology hub, with more than 18,000 ICT companies, nearly 360,000 technology-related workers, 24 percent of total wages tied to the technology sector, and 57 percent of services exports classified as technology-related.[12]

Bureau of Economic Analysis ("BEA") data show the same distinction in output terms. In the fourth quarter of 2025, Washington's information-sector GDP was approximately $146.7 billion, compared with approximately $910.1 billion for all Washington industries, or about 16.1 percent. Pennsylvania's information-sector GDP was approximately $45.2 billion, compared with approximately $1.081 trillion for all Pennsylvania industries, or about 4.2 percent.[13]

The comparison is also visible against manufacturing. In the fourth quarter of 2025, Washington's information-sector GDP was more than twice its manufacturing GDP, approximately $146.7 billion versus $68.0 billion. Pennsylvania showed the opposite pattern: manufacturing GDP was more than twice information-sector GDP, approximately $111.0 billion versus $45.2 billion. Pennsylvania economic-development materials describe a different mix, emphasizing agriculture, energy, life sciences, manufacturing, and robotics and technology as key sectors.[14]

These facts supply a plausible mechanism for further inquiry in certain cases, but they do not establish that AI caused Washington's unemployment pattern. A state with a larger information and technology footprint may show technology-related labor adjustments earlier or more visibly than a state with a different industrial mix. But the merger question still requires narrowing: which workers, which employers, which geography, which AI asset, and what but-for world? In an actual AI-related merger investigation, the Washington-Pennsylvania result should not be offered as a standalone allegation. It should guide evidence gathering.

## Conclusion

AI-related merger review requires evidentiary discipline. Agencies and courts should not assume that AI harms workers simply because it automates or alters tasks. They also should not assume that AI efficiencies eliminate labor concerns simply because the technology is innovative.

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[12] Washington State, Department of Commerce, *Information and Communication Technology in Washington State*—Choose Washington, https://choosewashingtonstate.com/why-washington/our-key-sectors/ict/.

[13] U.S. Bureau of Economic Analysis, Gross Domestic Product: Information (51) in Washington [WAINFONQGSP]; Gross Domestic Product: All Industry Total in Washington [WANQGSP]; Gross Domestic Product: Information (51) in Pennsylvania [PAINFONQGSP]; Gross Domestic Product: All Industry Total in Pennsylvania [PANQGSP], FRED, Federal Reserve Bank of St. Louis, Q4 2025 seasonally adjusted annual-rate data, updated Apr. 9, 2026. [Percentages are authors' calculations].

[14] Commonwealth of Pennsylvania, Key Industries, Pennsylvania Gets It Done / https://pagetsitdone.com/key_industries; U.S. Bureau of Economic Analysis, Gross Domestic Product: Manufacturing (31–33) in Washington [WAMANNQGSP] and Gross Domestic Product: Manufacturing (31–33) in Pennsylvania [PAMANNQGSP], FRED, Federal Reserve Bank of St. Louis, Q4 2025 seasonally adjusted annual-rate data, updated Apr. 9, 2026.

Section 7 requires a transaction-specific inquiry into whether the merger may substantially lessen competition in a relevant market, including relevant product markets and labor markets.

This article introduces BET as one way to think about that inquiry when the relationship between AI-related variables and labor outcomes is uncertain. BET is useful because it focuses attention on statistical dependence without requiring the analyst to impose a straight-line model, a particular data distribution, or a predetermined story about how AI must affect workers. That model-light advantage is valuable but not dispositive. BET does not define markets, prove causation, or replace merger-specific evidence.

BET's practical contribution is straightforward. It can help antitrust lawyers understand whether data show recurring, non-random dependence that may justify further inquiry. Public state-level evidence can illustrate the logic and generate questions. Firm-level, platform-level, occupation-level, and worksite-level evidence would be necessary to test any actual merger theory. Used in this focused way, BET can make AI-related labor-market analysis more disciplined.